KPK BOARD TEXTBOOK NOTES

Physics - Class 9

Unit 8: Thermal Properties of Matter

Complete notes in English, based on *A Textbook of Physics, Grade IX*
Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar

> **How to use these notes.** Read one section at a time, learn the green definition boxes by heart, and copy every yellow formula box into your notebook. Work through the solved examples and numericals before you try the multiple-choice questions (MCQs) at the end. The last page is a one-page revision sheet for the night before the exam.

What this unit covers

1. Temperature and heat - the kinetic theory explanation
2. Thermometers, thermometric properties and temperature scales
3. Conversion between Celsius, Fahrenheit and Kelvin scales
4. Thermal expansion of solids - linear and volume expansion
5. Thermal expansion of liquids - real and apparent expansion, and the anomalous expansion of water
6. Applications and consequences of thermal expansion
7. Heat capacity and specific heat capacity
8. Latent heat of fusion and latent heat of vaporization
9. Evaporation and cooling by evaporation

After studying this unit, you should be able to

- Describe the concept of temperature and heat in terms of the kinetic theory of matter.
- Describe the construction and use of a liquid-in-glass thermometer and the need for fixed points.
- Describe the structure of the Celsius, Fahrenheit and Kelvin scales and convert temperatures between them.
- Define thermal expansion, the coefficient of linear expansion and the coefficient of volume expansion, and solve related problems.
- Show the relationship between the coefficient of linear expansion and the coefficient of volume expansion.
- Define heat capacity and specific heat capacity, and explain the low mass density of ice using the anomalous expansion of water.
- Define latent heat of fusion and latent heat of vaporization, and describe evaporation and cooling by evaporation.

This checklist follows the Student Learning Outcomes for Unit 8 in the textbook.

8.1 Temperature

We say a body is hot or cold by touch, but touch is not reliable. Temperature gives us an exact, measurable meaning of hotness and coldness.

> **Temperature** is the degree of hotness or coldness of a body. It is the physical quantity which determines the direction of flow of heat.

Kinetic theory explanation. All matter is made of atoms and molecules which are in a state of continuous vibration. These molecules possess kinetic energy (K.E). The temperature of a body is the average kinetic energy of its molecules. The greater the average kinetic energy of the molecules, the higher is the temperature of the body.

Thermal equilibrium

When two bodies at different temperatures are placed in contact, heat flows from the hotter body to the colder body until both reach the same temperature. They are then in **thermal equilibrium** with each other. Heat is a form of energy which is transferred due to a temperature difference; it is never transferred from a cold body to a hot body on its own.

8.2 Heat

> **Heat** is a form of energy which transfers from one body to another body due to temperature difference between them. Heat is the total kinetic energy of the molecules of a body.

The SI unit of heat is the **joule (J)**, because heat is a form of energy. An older unit, the calorie, is still sometimes seen: 1 calorie = 4.186 J.

Temperature or heat - do not mix them up

Temperature	Heat
Degree of hotness or coldness of a body	Form of energy transferred due to temperature difference
Average kinetic energy of the molecules	Total kinetic energy of the molecules
Measured in degree Celsius, degree Fahrenheit or kelvin	Measured in joules
Measured with a thermometer	Measured with a calorimeter
Does not depend on the mass of the body	Depends on the mass of the body

A spark from a firework has a very high temperature but very little heat, because its mass is tiny. A bucket of warm water has a low temperature but stores much more heat energy. This example shows clearly why temperature and heat are different quantities.

8.3 Thermometer and Thermometry

The branch of physics which deals with the measurement of temperature is called **thermometry**. Our sense of touch cannot measure temperature accurately - the same lukewarm water feels warm to a hand that was in cold water and cool to a hand that was in hot water - so we need a reliable instrument.

> **Thermometer** is a device or instrument used to measure the temperature of a body accurately.

8.3.1 Thermometric property

> **Thermometric property** is a physical property of a substance that changes uniformly with temperature and can be used for the measurement of temperature.

Examples of thermometric properties include the volume of a liquid, the length of a solid, gas pressure, electrical resistance and electromotive force. The most commonly used thermometric property is the **thermal expansion of materials**: matter expands on heating and contracts on cooling, and the degree of expansion can be marked on a scale to record temperature.

Liquid-in-glass thermometer

The liquid-in-glass thermometer uses the variation in volume (length of the liquid thread) with temperature. The liquid is sealed in a glass bulb joined to a capillary tube with a scale etched on the stem. When the bulb is heated, the liquid expands and rises in the capillary; when it cools, the liquid contracts and falls.

A good thermometric liquid must:

- be easily seen in the tube;
- expand or contract rapidly, and by a large amount, over a wide range of temperature;
- expand uniformly;
- not stick to the inside of the tube, otherwise the reading will be too high when the temperature is falling.

The two liquids commonly used are **mercury** and **alcohol**. A clinical thermometer is a mercury thermometer with a constriction (kink) in the capillary which stops the mercury thread from falling back at once, so the doctor can read the patient's temperature after removing it. Its range is short, 35 °C to 42 °C, because human body temperature stays within these limits.

Mercury or Alcohol - Which Liquid, and Why Not Water?

Mercury	Alcohol
Opaque and shiny, so the thread is easy to see	Colourless, so it is dyed (usually red) to be seen
Does not stick to glass	Sticks slightly to glass
Expands uniformly	Expands more than mercury for the same rise
Freezes at -39 °C, so it cannot measure very low temperatures	Freezes at about -115 °C, so it can measure very low temperatures
Boils at 357 °C, so it can measure high temperatures	Boils at 78 °C, so it cannot measure high temperatures

Why is water not used in a thermometer? Water does not expand uniformly - between 0 °C and 4 °C it actually contracts on heating. It also sticks to glass, is colourless and hard to see, and its expansion is small. For these reasons water is unsuitable as a thermometric liquid.

8.3.2 Temperature Scales

A temperature scale is made from two reference points called **fixed points**: the **ice point** (lower fixed point), the temperature of pure melting ice, and the **steam point** (upper fixed point), the temperature of steam from pure boiling water, both at standard atmospheric pressure. The interval between the fixed points is divided into equal divisions.

Scale	Ice point	Steam point	Divisions	Used for
Celsius (centigrade), by Anders Celsius	0 °C	100 °C	100	Laboratory and daily life; SI-related work
Fahrenheit, by Daniel Gabriel Fahrenheit	32 °F	212 °F	180	Clinical (body) temperature
Kelvin (absolute), by William Thomson, Lord Kelvin	273 K	373 K	100	SI unit of temperature; science

> **Absolute zero** is the lowest possible temperature, at which the molecular movement of matter ceases. It is 0 K, which is equal to -273 °C. Kelvin temperatures are never negative.

Note that a *change* of 1 °C is exactly the same size as a change of 1 K, because both scales have 100 divisions between the same fixed points. Only the starting point (zero) is different.

8.3.3 Converting Between Temperature Scales

The same temperature can be written on any scale. The general rule compares how far the temperature is above the ice point with the total number of divisions between the fixed points. It gives the working formulas:

$$T_{°F} = (9/5) \times T_{°C} + 32 \quad | \quad T_{°C} = (5/9) \times (T_{°F} - 32)$$

$$T_K = T_{°C} + 273 \quad | \quad T_{°C} = T_K - 273$$

To convert Fahrenheit into Kelvin, first convert Fahrenheit into Celsius, then add 273. **Memory check:** -40 is the one temperature that reads the same on the Celsius and Fahrenheit scales: -40 °C = -40 °F.

Solved Example 8.1 - Human body temperature (from the book)

Question. The temperature of a normal human body is 37 °C. Find this temperature on the Fahrenheit and Kelvin scales.

Solution. Fahrenheit: $T_{°F} = (9/5)(37) + 32 = 66.6 + 32 =$ **98.6 °F**.

Kelvin: $T_K = 37 + 273 =$ **310 K**.

So 37 °C = 98.6 °F = 310 K.

Try the book assignment yourself - with the answer

Assignment 8.1. The temperature of an object is 250 K. Find its temperature in centigrade.

Answer. $T_{°C} = 250 - 273 =$ **-23 °C**.

Common mistake: Never write a degree sign with kelvin. We write 310 K, not 310 °K. Also remember that in the Fahrenheit formula the 32 is subtracted *before* multiplying by 5/9 when going from °F to °C.

8.4 Thermal Expansion

> **Thermal expansion** is the increase in size of a substance on heating. Most substances expand when heated and contract when cooled. For the same substance, gases expand more than liquids, and liquids expand more than solids.

Why does a solid expand? When the temperature of a solid is raised, its molecules vibrate through a larger distance. The increase in the amplitude of vibration increases the average distance between the molecules, so the solid expands. On cooling, the opposite happens and the solid contracts.

8.4.1 Linear thermal expansion of solids

The increase in length of a substance due to a rise in temperature is called **linear thermal expansion**. Experiments show that the change in length Δl is directly proportional to the original length l_o and to the change in temperature ΔT. A 4 m iron rod expands twice as much as a 2 m iron rod for the same rise in temperature.

$$\Delta l = \alpha \times l_o \times \Delta T \quad \text{and} \quad l_T = l_o (1 + \alpha \Delta T)$$

> **Coefficient of linear thermal expansion (α)** is the increase in length per unit length of a solid per kelvin (or per degree Celsius) rise in temperature: $\alpha = \Delta l / (l_o \Delta T)$. Its unit is K^{-1} (or $°C^{-1}$).

Table 8.2 - Thermal expansion coefficients at 20 °C (from the book)

Material	α (x 10^{-6} K^{-1})	γ (x 10^{-6} K^{-1})
Aluminium	25	75
Brass	19	57
Copper	17	51
Iron or steel	12	35
Lead	29	87
Concrete, brick	12	36
Ethyl alcohol (fluid)	-	1100
Petrol (fluid)	-	950
Mercury (fluid)	-	180
Water (fluid)	-	210
Air and most other gases	-	3400

Solved Example 8.2 - Bridge Engineering (from the book)

Question. The steel bed of a suspension bridge is 200 m long at 20 °C ($\alpha = 12 \times 10^{-6}\ K^{-1}$). If the extremes of temperature to which it might be exposed are -5 °C to 50 °C, how much will it contract and expand?

Solution (a) Contraction. $\Delta T_1 = -5 - 20 = -25$ K. $\Delta l_1 = \alpha\, l_o\, \Delta T_1 = 12 \times 10^{-6} \times 200 \times (-25) = -6 \times 10^{-2}$ m = **-6 cm** (it contracts by 6 cm).

Solution (b) Expansion. $\Delta T_2 = 50 - 20 = 30$ K. $\Delta l_2 = 12 \times 10^{-6} \times 200 \times 30 = 7.2 \times 10^{-2}$ m = **7.2 cm**.

The total range the expansion joints must accommodate is 7.2 cm + 6 cm = **13.2 cm**.

Book Assignment 8.2 for practice: a metal bar 60 cm long is heated from 8 °C to 100 °C and its length becomes 60.127 cm. Its coefficient of linear expansion is $\alpha = 0.127 / (60 \times 92) = 2.3 \times 10^{-5}\ K^{-1}$.

Volume (cubical) thermal expansion of solids

The increase in volume of a substance due to a rise in temperature is called **volume thermal expansion**. The change in volume is directly proportional to the original volume V_o and to the rise in temperature:

$$\Delta V = \gamma \times V_o \times \Delta T \quad \text{and} \quad V_T = V_o\,(1 + \gamma\,\Delta T)$$

Coefficient of volume thermal expansion (γ) is the change in volume per unit volume per kelvin change in temperature: $\gamma = \Delta V / (V_o\,\Delta T)$. Its unit is K^{-1}.

Solids expand to the same extent in all three directions, so the volume coefficient is about three times the linear coefficient:

$$\gamma = 3\alpha$$

Solved Example 8.3 - Glass laser rod (from the book)

Question. A glass rod is 30.0 cm long and has volume $5.30 \times 10^{-5}\ m^3$. Take $\alpha = 9.00 \times 10^{-6}\ K^{-1}$. If its temperature increases by 65.0 °C, find the increase in (a) its length and (b) its volume.

Solution (a) $\Delta l = \alpha\, l_o\, \Delta T = 9.00 \times 10^{-6} \times 0.300 \times 65 =$ **1.76×10^{-4} m**.

Solution (b) $\gamma = 3\alpha$, so $\Delta V = 3\alpha\, V_o\, \Delta T = 3 \times 9.00 \times 10^{-6} \times 5.30 \times 10^{-5} \times 65 =$ **$93.0 \times 10^{-9}\ m^3$**.

8.4.2 Thermal Expansion of Liquids

Like solids, liquids also expand on heating and contract on cooling. A liquid has no definite length or surface area of its own - it takes the shape of its container - so for liquids we consider **only volume expansion**. Heat affects both the liquid and its container, so the true (real) expansion of a liquid cannot be observed directly. Liquids therefore have two kinds of expansion:

> **Real expansion (V_R)** is the actual increase in the volume of a liquid due to a rise in temperature, independent of the expansion of the container.

> **Apparent expansion (V_A)** is the observed increase in the volume of a liquid. It is the real expansion minus the expansion of the container: $V_A = V_R - V_C$.

When a flask of liquid is heated, the vessel expands first and the liquid level falls a little (level A to B). Then the liquid itself expands and rises past its original level (to C). So:

$$\text{Real expansion} = \text{Apparent expansion} + \text{Expansion of the vessel}$$
$$\gamma_R = \gamma_A + \gamma_{\text{vessel}}$$

where γ_R = real increase in volume / (original volume x rise in temperature) and γ_A = apparent increase in volume / (original volume x rise in temperature). Both coefficients have unit K^{-1}.

Anomalous expansion of water

Liquids expand on heating - except water between 0 °C and 4 °C. When water at 0 °C is heated, its volume *decreases* up to 4 °C, and only above 4 °C does its volume increase. This peculiar behaviour, caused by the formation of hydrogen bonds, is called the **anomalous expansion of water**.

- From 0 °C to 4 °C the density of water increases; above 4 °C the density decreases.
- Water therefore has **maximum density at 4 °C**.
- Because ice is less dense than water, ice floats - ice cubes float in a drink and icebergs float in the ocean.
- In winter the water at the bottom of a lake stays at 4 °C while ice forms on top, so fish and other aquatic life survive under the ice.

Solved Example 8.4 - Lost Diesel (from the book)

Question. An oil trucker loaded 37,000 L of diesel fuel ($\gamma = 9.5 \times 10^{-4}\ K^{-1}$) at Jacobabad, where on a hot day the temperature is 50 °C. The oil is transported to Kalam, where on a cold day the temperature is 0 °C. What is the change in volume of the diesel? How many litres did he deliver?

Solution. $\Delta T = 0 - 50 = -50$ K. $\Delta V = \gamma V_0 \Delta T = 9.5 \times 10^{-4} \times 37{,}000 \times (-50) = -1757.5$ L, about **-1760 L**.

Volume delivered = 37,000 - 1760 = **35,240 L**. The diesel did not leak - it contracted because it cooled.

Book Assignment 8.4 for practice: petrol at 0 °C occupies 250 L; at 50 °C ($\gamma = 9.6 \times 10^{-4}\ K^{-1}$) its volume is $V = 250\ (1 + 9.6 \times 10^{-4} \times 50) = 262$ L.

8.4.3 Applications and Consequences of Thermal Expansion

1. **Railway lines.** Engineers leave a small gap between two rails. Without gaps, expanding rails would push against each other, bend, or break free, causing derailment and major accidents.
2. **Opening a tight jar lid.** Holding a tight metal lid under hot water loosens it, because metal expands more than glass for the same temperature change.
3. **Transmission lines.** Overhead wires are hung loosely because they sag more in summer, when they expand, than in winter.
4. **Shrink fitting.** An axle is cooled in liquid nitrogen at -196 °C so it shrinks and the gear wheel slips on; back at normal temperature the axle expands to give a very tight fit.
5. **Iron ring on a cart wheel.** The iron ring, slightly smaller than the wooden wheel, is heated so it expands and fits over the wheel; on cooling it contracts into a tight fit.
6. **Expansion joints in bridges.** Large bridges have comb-like joints whose teeth move together as the bridge expands in the day's heat and apart as it cools at night, giving the bridge a built-in safety zone.
7. **Bimetallic strip.** Two strips of different metals (for example copper and iron) riveted together bend on heating because the metals expand by different amounts - copper expands more, so it stays on the outside of the bend. Bimetallic strips are used in fire alarms and thermostats (for example in an electric iron) to switch a circuit on or off automatically.

Do holes expand or contract? A hole in a heated metal plate gets *larger*, not smaller. Every part of the material expands in the same proportion, exactly as if the material inside the hole were still present and expanding too.

8.5 Heat Capacity and Specific Heat Capacity

When a substance absorbs heat, its temperature rises; when it loses heat, its temperature falls. Different materials respond differently: water needs a lot of heat to change its temperature, while silver needs very little. A massive body also needs more heat than a lighter body of the same material for the same temperature rise.

8.5.1 Heat capacity (thermal capacity)

> **Heat capacity (C_m)** of a substance is the quantity of heat required to raise the temperature of that mass of the substance by 1 °C or 1 K: $C_m = \Delta Q / \Delta T$. Its SI unit is joule per kelvin, J K^{-1}.

The value of heat capacity depends on (1) the nature of the material, (2) the mass of the material, and (3) the rise in temperature considered.

8.5.2 Specific heat capacity (specific heat)

Heat equal masses of copper, iron and water under the same flame for the same time. They absorb the same heat, but the rise in temperature is maximum for copper, minimum for water and in between for iron. Copper heats up quickly; water heats up very slowly.

> **Specific heat capacity (c)** is the quantity of heat required to raise the temperature of unit mass (1 kg) of a substance by 1 °C or 1 K: $c = \Delta Q / (m \Delta T)$. Its SI unit is J kg^{-1} K^{-1}.

$$\Delta Q = m \times c \times \Delta T$$

Table 8.3 - Specific heats of various substances (from the book)

Material	c (J kg^{-1} K^{-1})	Material	c (J kg^{-1} K^{-1})
Water	4190	Aluminium	910
Ethanol	2450	Glass	840
Glycerin	2410	Air	721
Wood	1700	Oxygen	651
Steam (100 °C)	1520	Carbon dioxide	638
Iron / steel	452	Copper	387
Silver	230	Mercury	139
Gold	129	Lead	128

Raising the temperature of 1 kg of water by 1 K needs 4190 J, but the same rise for 1 kg of aluminium needs only 910 J. The specific heat of water (4190 J kg^{-1} K^{-1} at 15 °C) is higher than that of any other common substance, and this has natural benefits in sustaining life on Earth.

8.5.3 Importance of the High Specific Heat of Water

1. **Moderate climate of the sea shore.** The specific heat of sand is about 800 J kg^{-1} K^{-1}, so water needs about five times more heat than the same mass of soil for a 1 K rise. Land heats up and cools down much faster than the sea, and this difference produces land breeze and sea breeze. It keeps the climate of coastal areas moderate; the monsoon in Pakistan is also due to this land-sea temperature difference.
2. **As a coolant.** Water flowing through the radiator pipes of a vehicle removes heat from the hot parts. It extracts much heat without much rise in its own temperature.
3. **Hot water bottles and heating.** Water stays hot for a long time because it gives out a large amount of heat for a small fall in temperature, so it is used in hot water bottles and central heating systems.
4. **Farming.** Farmers fill their fields with water on a frosty night; the water releases heat slowly and protects the crop.

8.6 Latent Heat

When ice is heated, its temperature rises to 0 °C and then stays at 0 °C while the ice melts - the heat supplied changes the state, not the temperature. The same happens when water boils at 100 °C. This hidden heat is called latent heat (latent means hidden).

> **Latent heat** is the heat energy required to change the state of a substance without changing its temperature.

Latent heat of fusion

> **Specific latent heat of fusion (L_f)** is the heat required to melt unit mass (1 kg) of a solid completely into liquid at its melting point, without change in temperature. For ice, $L_f = 3.36 \times 10^5$ J kg^{-1}.

$$Q = m \times L_f$$

This is why ice at 0 °C cools a drink better than the same mass of water at 0 °C: the ice first absorbs 3.36 x 10^5 J per kilogram as latent heat just to melt, and only then starts warming up.

Latent heat of vaporization

> **Specific latent heat of vaporization (L_v)** is the heat required to convert unit mass (1 kg) of a liquid completely into vapour at its boiling point, without change in temperature. For water, $L_v = 2.26 \times 10^6$ J kg^{-1} (22.6×10^5 J kg^{-1}).

$$Q = m \times L_v$$

This is why steam burns are more severe than boiling-water burns: steam first releases its large latent heat (2.26 x 10^6 J per kilogram) as it condenses on the skin, and then the hot water formed cools down, releasing still more heat.

The Heating Curve - Reading the Plateaus

If ice at -10 °C is heated steadily and its temperature is plotted against time, the graph rises, then becomes flat while the ice melts at 0 °C, rises again through liquid water, becomes flat again while the water boils at 100 °C, and finally rises as steam. The two flat parts (plateaus) are where latent heat is being absorbed: energy is separating the molecules instead of speeding them up, so the temperature does not change.

8.7 Evaporation

> **Evaporation** is the change of a liquid into vapour at any temperature, taking place from the surface of the liquid only. It is a slow and silent process.

Kinetic theory explanation. The molecules of a liquid move with different speeds. The fastest molecules near the surface escape into the air, leaving behind the slower, cooler molecules. The average kinetic energy - and therefore the temperature - of the remaining liquid falls. That is why evaporation produces cooling: the liquid takes the latent heat of vaporization from its surroundings.

Evaporation or boiling?

Evaporation	Boiling
Occurs at all temperatures	Occurs only at the boiling point
Occurs only at the surface of the liquid	Occurs throughout the liquid (bubbles form in the bulk)
Slow and silent process	Fast and violent process
Temperature of the liquid falls (cooling)	Temperature stays constant at the boiling point
Rate depends on surface conditions	Rate depends on the heat supplied

Factors affecting the rate of evaporation

1. **Temperature:** evaporation is faster at higher temperature (clothes dry quickly on a hot day).
2. **Surface area:** evaporation is faster from a larger exposed surface (tea cools faster in a saucer than in a cup).
3. **Wind:** moving air carries the vapour away, so evaporation is faster in wind.
4. **Humidity:** evaporation is slower when the air is already humid, and faster in dry air.
5. **Nature of the liquid:** volatile liquids such as petrol and ether evaporate faster than water.

Cooling by Evaporation - Applications

1. **Earthen (clay) water pots** keep water cool because water seeps through their tiny pores and evaporates from the outer surface, taking heat from the water inside.
2. **Desert coolers** work on hot, dry days: water evaporates from wet pads and cools the air blown through them. They fail on humid days because evaporation becomes slow.
3. **Sprinkling water** on a hot roof or courtyard cools it, because the water takes latent heat from the surface as it evaporates.
4. **Cotton clothes in summer** absorb sweat, and the evaporation of the sweat cools the body.
5. **Sweating and panting:** evaporation of sweat cools our skin; dogs, which have almost no sweat glands on the body, hang out their tongues and pant so that evaporation from the tongue cools them.
6. **Refrigerators and air conditioners** use evaporation of a special liquid (refrigerant) inside pipes to absorb heat from the cabinet or room.

Solved Numericals - From the Book Exercise

N1. Convert 90 °F into Celsius. $T_{°C}$ = (5/9)(90 - 32) = (5/9)(58) = **32.2 °C**.

N2. Convert 220 K into Fahrenheit. $T_{°C}$ = 220 - 273 = -53 °C. $T_{°F}$ = (9/5)(-53) + 32 = -95.4 + 32 = **-63.4 °F**.

N3. A steel rod is 50 cm long at 0 °C. What is its length at 100 °C? (α for steel = 12 x 10^{-6} K^{-1}.) Δl = 12 x 10^{-6} x 0.50 x 100 = 6.0 x 10^{-4} m = 0.06 cm. Length = **50.06 cm**.

N4. A brass rod of length 2.5 m is heated and its temperature rises by 40 K. Find the increase in length. (α for brass = 1.8 x 10^{-5} K^{-1}.) Δl = 1.8 x 10^{-5} x 2.5 x 40 = 1.8 x 10^{-3} m = **1.8 mm**.

N5. A metal rod is 200 cm long at 30 °C and 200.32 cm long at 130 °C. Find α. α = Δl / (l_o ΔT) = 0.32 / (200 x 100) = **1.6 x 10^{-5} K^{-1}**.

N6. A Pyrex beaker contains 200 cm^3 of mercury at 20 °C. It is heated to 100 °C. How much mercury overflows? (γ mercury = 1.8 x 10^{-4} K^{-1}, γ Pyrex = 0.3 x 10^{-4} K^{-1}.) Overflow is the apparent expansion: ΔV_A = (1.8 - 0.3) x 10^{-4} x 200 x 80 = **2.4 cm^3**.

N7. How much heat is needed to melt 200 g of ice at 0 °C into water at 0 °C? (L_f = 3.36 x 10^5 J kg^{-1}.) Q = mL_f = 0.200 x 3.36 x 10^5 = **67,200 J**.

Short Questions - Book Concepts, Checked Answers

Q1. Why does a clinical thermometer have a short range? Because it is made only to measure human body temperature, which remains between 35 °C and 42 °C even in severe fever or hypothermia. A wider scale is unnecessary and would be less accurate to read.

Q2. Mercury is used in thermometers even though it is poisonous. Why? Because of its useful properties: it is opaque and shiny (easy to see), expands uniformly, does not stick to glass, is a good conductor of heat (responds quickly), and stays liquid over a wide range, from -39 °C to 357 °C.

Q3. Where is alcohol better than mercury? Alcohol freezes at about -115 °C, so alcohol thermometers can measure very low temperatures that mercury cannot. Alcohol also expands more per degree and can be coloured for easy reading, but it boils at 78 °C and sticks slightly to glass.

Q4. Why do steam burns hurt more than boiling-water burns? Each gram of steam gives out its latent heat of vaporization (2260 J per gram) as it condenses on the skin, in addition to the heat the hot water then releases while cooling. Boiling water gives out only the second part.

Q5. Why does a desert cooler cool better on a hot, dry day? Evaporation is faster when the air is hot and dry. On a humid day the air already holds much water vapour, evaporation slows down, and the cooler stops cooling effectively.

Q6. Why does ice float on water? Because of the anomalous expansion of water: water has maximum density at 4 °C, and ice is less dense than liquid water, so it floats. The same property lets lakes freeze from the top while liquid water at 4 °C remains at the bottom.

Q7. On a hot day, why do we sprinkle water on the roof? The sprinkled water evaporates and takes its latent heat of vaporization from the roof surface, cooling the roof and the rooms below.

MCQs - Choose the Correct Answer

1. The steam point on the Kelvin scale is: (a) 0 K (b) 100 K (c) 273 K (d) 373 K
2. The range of a clinical thermometer is: (a) -10 °C to 110 °C (b) 35 °C to 42 °C (c) 0 °C to 100 °C (d) -35 °C to 42 °C
3. The coefficient of linear expansion of brass is: (a) $1.9 \times 10^{-5}\ K^{-1}$ (b) $1.9 \times 10^{-3}\ K^{-1}$ (c) $5.7 \times 10^{-5}\ K^{-1}$ (d) $1.2 \times 10^{-5}\ K^{-1}$
4. For a solid, the coefficient of volume expansion γ is related to α by: (a) γ = α (b) γ = 2α (c) γ = 3α (d) γ = α/3
5. Mercury freezes at -39 °C, so a mercury thermometer cannot be used to measure: (a) 0 °C (b) -30 °C (c) -50 °C (d) 37 °C
6. Absolute zero is equal to: (a) 0 °C (b) -273 °C (c) 273 K (d) -100 °C
7. The specific heat of water is: (a) 910 J $kg^{-1}\ K^{-1}$ (b) 2450 J $kg^{-1}\ K^{-1}$ (c) 4190 J $kg^{-1}\ K^{-1}$ (d) 139 J $kg^{-1}\ K^{-1}$
8. The specific latent heat of fusion of ice is: (a) 3.36×10^{5} J kg^{-1} (b) 2.26×10^{6} J kg^{-1} (c) 4.19×10^{3} J kg^{-1} (d) 1.8×10^{-4} J kg^{-1}
9. Water has its maximum density at: (a) 0 °C (b) 4 °C (c) 100 °C (d) -4 °C
10. Evaporation takes place: (a) only at the boiling point (b) from the whole liquid (c) at all temperatures, from the surface only (d) only in summer

> **Answer key:** 1 (d) 2 (b) 3 (a) 4 (c) 5 (c) 6 (b) 7 (c) 8 (a) 9 (b) 10 (c)

One-Page Revision Sheet

Definitions to learn by heart

- **Temperature:** average kinetic energy of the molecules; degree of hotness or coldness.
- **Heat:** total kinetic energy of the molecules, transferred due to temperature difference; unit joule.
- **Thermometric property:** a property that changes uniformly with temperature, used to measure it.
- **Absolute zero:** 0 K = -273 °C, where molecular motion ceases.
- **Thermal expansion:** increase in size of a substance on heating.
- **α:** increase in length per unit length per kelvin rise. **γ:** change in volume per unit volume per kelvin; γ = 3α for solids.
- **Heat capacity:** heat to raise the temperature of the whole mass by 1 K (J K^{-1}).
- **Specific heat (c):** heat to raise 1 kg by 1 K (J kg^{-1} K^{-1}); water = 4190, the highest common value.
- **Latent heat:** heat that changes state without changing temperature; L_f (ice) = 3.36 x 10^5 J kg^{-1}; L_v (water) = 2.26 x 10^6 J kg^{-1}.
- **Evaporation:** surface change of liquid into vapour at all temperatures; it produces cooling.

Formula bank

$$T_{°F} = (9/5)\,T_{°C} + 32 \quad | \quad T_{°C} = (5/9)(T_{°F} - 32) \quad | \quad T_K = T_{°C} + 273$$

$$\Delta l = \alpha\, l_o\, \Delta T \quad | \quad \Delta V = \gamma\, V_o\, \Delta T \quad | \quad \gamma = 3\alpha \quad | \quad \gamma_R = \gamma_A + \gamma_{vessel}$$

$$\Delta Q = m\, c\, \Delta T \quad | \quad Q = m\, L_f \quad | \quad Q = m\, L_v$$

Fixed points at a glance

Scale	Ice point	Steam point	Divisions
Celsius	0 °C	100 °C	100
Fahrenheit	32 °F	212 °F	180
Kelvin	273 K	373 K	100

Five facts examiners love

1. 37 °C = 98.6 °F = 310 K (normal body temperature).
2. Water contracts from 0 °C to 4 °C and has maximum density at 4 °C - so ice floats.
3. A hole in a heated plate gets bigger, and a 1 °C change equals a 1 K change.
4. Real expansion of a liquid is always greater than its apparent expansion.
5. Evaporation is faster with higher temperature, larger surface area, more wind and lower humidity.

End of Unit 8 notes. Source textbook: A Textbook of Physics, Grade IX, Khyber Pakhtunkhwa Textbook Board, Peshawar (book pp. 197-220). Prepared by Shah Fahad - 03409090945.